

for fireworks, it should not experience sudden negative acceleration after zero.

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$$\begin{aligned}v_y &= u_y + at \\&= -8 + 1.62(9) \\&= 6.58 \text{ ms}^{-1}\end{aligned}$$

$$\begin{aligned}\text{speed} &= \sqrt{v_y^2 + v_x^2} \\&= \sqrt{6.58^2 + 4^2} \\&= 7.7 \text{ ms}^{-1}\end{aligned}$$

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